

Anomalous Phase Behavior of Polymer–Ionic Liquid Mixtures

Published as part of The Journal of Physical Chemistry B special issue “Athanasios Z. Panagiotopoulos Festschrift”.

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Cite This: *J. Phys. Chem. B* 2025, 129, 12032–12041



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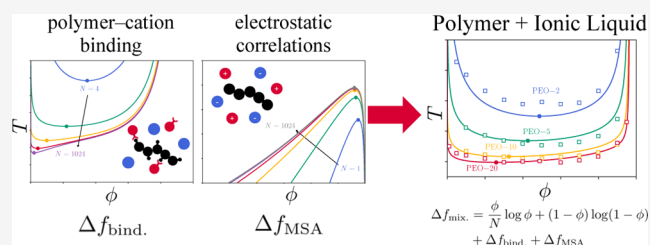


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ABSTRACT: Polymer–ionic liquid (IL) mixtures are promising materials for applications ranging from battery electrolytes and gas separation membranes to responsive actuators and sensors, owing to their unique combination of mechanical robustness and ionic conductivity. These systems often exhibit lower critical solution temperature (LCST) behavior and more symmetric phase envelopes than anticipated from the classical Flory–Huggins theory. In this work, we investigate the molecular origins of this anomalous behavior by systematically incorporating clustering, polymer–cation binding and electrostatic correlations into a simple thermodynamic framework. Our analysis reveals that the formation of clusters contributes to demixing at low temperatures, contrary to both experimental observations and prior theoretical predictions. In contrast, electrostatic interactions introduce persistent asymmetries in the free energy landscape, yielding different chain-length scaling in the critical composition and interaction parameter than predicted by the Flory–Huggins theory. Based on these insights, we propose a minimal, physically grounded model incorporating only mixing entropy, temperature-dependent Flory–Huggins parameter, and electrostatic correlation. This model successfully reproduces key experimental features in systems such as PEO + [EMIM][BF₄], using a small set of interpretable parameters. These results offer a mechanistic understanding of the phase behavior of polymer–IL mixtures and provide a practical modeling framework for their design and optimization.



1. INTRODUCTION

Polymer–ionic liquid (IL) mixtures are promising materials for a range of applications. Owing to their thermal and chemical stability, nonvolatility, and high ionic conductivity, ionic liquids have been widely explored for use in battery electrolytes.^{1–3} When combined with polymers, which impart mechanical strength and structural integrity, these mixtures yield durable and efficient electrolytes.^{4–6} Beyond energy storage, polymer–IL systems have also shown utility in gas and ion separation membranes,^{7–10} soft actuators,^{11,12} and chemical sensors.^{13–15} Despite this versatility, the phase behavior of these mixtures remains poorly understood.¹⁶

At first glance, polymer–IL mixtures appear to resemble conventional polymer solutions and should therefore be well described by the classical Flory–Huggins theory,^{17,18} with the familiar expression for the free energy of mixing ΔF_{mix}

$$\frac{\nu \Delta F_{\text{mix}}}{k_B T V} = \frac{\phi}{N} \log \phi + (1 - \phi) \log(1 - \phi) + \chi \phi(1 - \phi) \quad (1)$$

In eq 1, ν is the monomer volume (assumed to be the same as that of the solvent), V is the system volume, T the temperature, k_B the Boltzmann constant, N the chain length of the polymer, ϕ the polymer volume fraction and χ the Flory–Huggins parameter characterizing the polymer–solvent interaction.

This expression has been extensively used to model polymer solutions and blends,^{19,20} often with χ taken to be temperature-dependent. Notably, it predicts that the critical volume fraction and interaction parameter scale as $\phi_c = N^{-1/2}$ and $\chi_c = \frac{1}{2} + N^{-1/2}$, respectively, for large N .

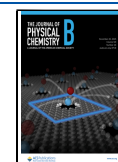
However, experiments by Lee et al.^{21,22} with mixtures of poly(ethylene oxide) (PEO) and 1-ethyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF₄]) revealed anomalous behavior that is very different from that anticipated from Flory–Huggins theory. Given the hydrophilic nature of PEO and the relatively hydrophobic character of [EMIM][BF₄], one would expect a positive interaction parameter χ that decreases with temperature, leading to upper critical solution temperature (UCST) behavior. Additionally, due to size asymmetry, the critical concentration would be expected to vanish as $N \rightarrow \infty$. Instead, Lee et al. observed lower critical solution temperature (LCST) phase separation with a critical composition near $\phi_c \approx$

Received: July 13, 2025

Revised: October 6, 2025

Accepted: October 30, 2025

Published: November 6, 2025



0.5, relatively insensitive to chain length. This trend has been observed in other polymer-IL systems.^{23–29}

To explain this behavior, several approaches have been proposed. Molecular dynamics simulations^{30,31} have successfully reproduced the experimental trends and offered valuable insights into the phase behavior. For instance, Choi et al.³⁰ emphasized the importance of polymer-cation binding, particularly between the ether oxygens on PEO and the electropositive hydrogens on the imidazolium ring of EMIM⁺. However, because molecular dynamics inherently capture all interactions simultaneously, it can be difficult to disentangle the specific contribution of each. In this context, theoretical approaches offer a clear advantage by enabling controlled, systematic investigation of individual interactions.

White and Lipson³² were the first to develop a theoretical approach to model polymer-IL systems. They used a lattice-based model incorporating the strong cohesive energy of ILs and their tendency to form long-lived clusters.^{33–35} By making the simplifying assumption that ILs form clusters of fixed size, their model successfully reproduced the more-symmetric shape of the phase envelope.

Another major simplification in the work of White and Lipson, and in others,^{36,37} is the treatment of electrostatic interactions. These studies typically assume that electrostatics are effectively captured through ion pairing or, in White and Lipson's case, through the formation of fixed-size clusters. Other approaches, such as that of Nakamura,³⁸ incorporate electrostatic effects via solvation free energies and dielectric mismatch. However, to our knowledge, no existing theoretical model has explicitly included electrostatic correlations, which are inherently nonpairwise, in polymer-IL mixtures using even basic frameworks such as the Debye–Hückel theory,³⁹ let alone more advanced treatments such as the mean spherical approximation (MSA)⁴⁰ or the hypernetted-chain (HNC) approach.⁴¹

Prior work on charged polymer systems by Sing and co-workers^{42,43} has shown that including electrostatic interactions can significantly shift phase boundaries relative to neutral analogs, arising from the many-body, long-range nature of these interactions in contrast to typical solvent–polymer interactions. This suggests that the same may hold for polymer-IL systems. We therefore hypothesize that electrostatic correlations are essential to understanding the anomalous phase behavior of polymer-IL mixtures.

To test this hypothesis, we systematically examine the role of clustering, polymer–cation binding and electrostatics in determining the phase behavior. Based on this analysis, we propose a minimal model that incorporates only the essential physics yet captures the experimentally observed trends. The remainder of this article is organized as follows: Sections 2 and 3 present the theoretical framework, including treatments of clustering, polymer–cation binding and electrostatic interactions, as well as the methods used to solve for the coexistence curve. Section 4 analyzes the influence of these contributions on the phase behavior, after which we propose a minimal, physically grounded model for polymer-IL mixtures in Section 5. Finally, Section 6 summarizes the key findings and implications of this work.

2. THEORY

To model the phase behavior of polymer–ionic liquid mixtures, we extend the Flory–Huggins (FH) theory, taking inspiration from the work by Jiang and co-workers^{44,45} (later extended by Zhang et al.⁴⁶). We express this theory in terms of the Helmholtz

free energy of mixing, ΔF_{mix} , as a sum of ideal and excess contributions:

$$\frac{\Delta F_{\text{mix}}}{k_{\text{B}}TV} = \Delta f_{\text{mix}} = \Delta f_{\text{id}} + \Delta f_{\text{ex}} \quad (2)$$

where T is the temperature, k_{B} is the Boltzmann constant and V is the system volume. The ideal mixing contribution, Δf_{id} , captures the translational entropy and is formally given by

$$\Delta f_{\text{id}} = \frac{1}{v} \left[\frac{\phi}{N} \log \phi + (1 - \phi) \log(1 - \phi) \right] \quad (3)$$

where ϕ and N are the volume fraction and degree of polymerization of the polymer and v is the reference monomer volume.

The excess contribution, Δf_{ex} , accounting for all other nonideal interactions, will depend on the system and the model representation. The simplest and most familiar treatment would be through the Flory–Huggins expression,^{17,47} in which the interactions between pairs of species are represented by the parameter χ

$$\Delta f_{\text{ex}} = \frac{1}{v} \chi \phi (1 - \phi) \quad (4)$$

A common extension introduces temperature dependence into χ , enabling the description of both UCST and LCST behavior.^{19,48}

$$\chi = A + \frac{B}{T} \quad (5)$$

where A and B are the entropic and enthalpic components, respectively. Note that the specific temperature dependence of the χ parameter does not alter the scaling with chain length of the critical properties. It is more rigorous, and in many cases necessary, to explicitly account for the underlying physical interactions that give rise to the phase behavior. In this work, we consider three such interactions explicitly: clustering, polymer–cation binding, and electrostatic correlations.

2.1. Clustering. A distinguishing feature of ionic liquids, relative to inorganic salts, is their tendency to form large and long-lived aggregates or clusters.^{33–35} This behavior has been proposed as a key factor underlying the atypical phase behavior of certain polymer-ionic liquid mixtures. For instance, White and Lipson³² suggested that clustering, with a characteristic aggregate size of ten, contributed to the unusual phase behavior of PEO + [EMIM][BF₄] mixtures. Several theoretical frameworks have been developed to describe clustering in these systems.^{49,50}

Building on Wertheim's thermodynamic perturbation theory (TPT1) and in a similar vein as the approach of McEldrew and co-workers,^{49,50} we adopt a simplified clustering model in which the cation and anion possess m_+ and m_- binding sites, respectively. The excess free energy contribution due to clustering in the mixture is given by (see the derivation in the Supporting Information)

$$v\Delta f_{\text{cl}} = \frac{1}{2}(1 - \phi) \left\{ m_+ \left[\log \left(\frac{X_+}{X_{+0}} \right) + \frac{1}{2}(X_{+0} - X_+) \right] + m_- \left[\log \left(\frac{X_-}{X_{-0}} \right) + \frac{1}{2}(X_{-0} - X_-) \right] \right\} \quad (6)$$

where X_i and X_{i0} denote the fractions of unbound sites of species i (or equivalently, the fraction of free species i) in the mixed and pure ionic liquid, respectively. These values are obtained self-consistently from the mass action equations

$$X_+ = \frac{1}{1 + \frac{1}{2}m_-(1 - \phi)\lambda_{cl}X_-} \quad (7)$$

$$X_- = \frac{1}{1 + \frac{1}{2}m_+(1 - \phi)\lambda_{cl}X_+} \quad (8)$$

Note that identical expressions can be obtained following the approach by Semenov and Rubinstein.⁵¹ The parameter λ_{cl} is the binding strength between cation and anion sites, defined as

$$\lambda_{cl} = v_{cl}\exp(\epsilon_{cl}/(k_B T)) \quad (9)$$

where v_{cl} is a dimensionless binding volume (normalized by the monomer volume), reflecting the steric accessibility, and ϵ_{cl} is the pairwise binding energy.⁵¹

From this framework, the average cluster size, $\langle N_c \rangle$, can be calculated as^{52,53}

$$\langle N_c \rangle = 1 + \frac{p_{+-}p_{-+}[(m_+ - 1)p_{+-} + (m_- - 1)p_{-+}]}{\left(\frac{p_{+-}}{m_-} + \frac{p_{-+}}{m_+}\right)[1 - (m_+ - 1)(m_- - 1)p_{+-}p_{-+}]} \quad (10)$$

where p_{+-} and p_{-+} denote the probabilities that a site on a cation is bound to an anion, and vice versa

$$p_{+-} = \frac{1}{2}(1 - \phi)m_-\lambda_{cl}X_+X_- \quad (11)$$

$$p_{-+} = \frac{1}{2}(1 - \phi)m_+\lambda_{cl}X_+X_- \quad (12)$$

From eq 10, it follows that percolation (i.e., the formation of infinite clusters) occurs when $1 - (m_+ - 1)(m_- - 1)(p_{+-}p_{-+}) = 0$, which can only be satisfied when $m_+ \geq 2$ and $m_- \geq 2$. For instance, when $m_+ = m_- = 2$, infinite clusters only arise in the limit $p_{+-} = p_{-+} = 1$, corresponding to infinite binding strength and chain-like assembly. In the case of $m_+ > 2$ and/or $m_- > 2$, percolation can occur at a finite binding strength. If either species has only a single binding site ($m_i = 1$), then the maximum cluster size is finite and limited to $m_+ + m_-$.

2.2. Polymer–Cation Binding. In their study, Lee et al.²² proposed (and later verified by Choi et al.³⁰) that cations such as EMIM⁺ can bind to the oxygen atoms in PEO via hydrogen bonding interactions from the imidazolium ring. This phenomenon is very similar to the hydrogen bonding interactions in water + PEO systems^{54,55} where the introduction of PEO results in entropy loss as water is now bound to the PEO backbone. Such interactions can be modeled analogously to the clustering framework described above, by introducing binding sites on both the polymer and the cation.

For simplicity, we assume one binding site per polymer repeat unit and one per cation. Under these assumptions, the excess free energy contribution from polymer–cation binding is given by

$$v\Delta f_b = \frac{1}{2}(1 - \phi)\left(\log X_+ + \frac{1 - X_+}{2}\right) + \phi\left(\log X_p + \frac{1 - X_p}{2}\right) \quad (13)$$

where X_+ and X_p are the fractions of unbound sites on the cation and polymer, respectively. Since binding is only considered between polymer and cation, the reference values $X_{i0} = 1$ for all species. The unbound site fractions are determined self-consistently from the following mass-action equations

$$X_+ = \frac{1}{1 + \phi\lambda_bX_p} \quad (14)$$

$$X_p = \frac{1}{1 + \frac{1}{2}m_+(1 - \phi)\lambda_bX_+} \quad (15)$$

where λ_b is the polymer–cation binding strength, defined analogously to eq 9.

While this model can, in principle, be extended to include both clustering and polymer–cation binding within a unified framework,⁴⁹ doing so requires solving a larger coupled system of equations and introduces additional parameters. To isolate the effects of each interaction and maintain interpretability, we consider clustering and binding contributions separately in this work.

2.3. Electrostatic Correlations. To model electrostatic interactions, several theoretical approaches are available, including the Debye–Hückel theory³⁹ and the hypernetted-chain approximation.⁴¹ Consistent with our previous work,^{46,48,56} we employ the mean spherical approximation (MSA), as developed by Blum and co-workers,^{40,57} which provides a favorable balance between analytical tractability and quantitative accuracy. The generalized form of the MSA free energy requires solving a self-consistent set of equations. However, in the special case of equal size for all ions, the free energy can be simplified to give an analytical expression

$$\Delta f_{ele} = \frac{1}{\pi}\left[(1 - \phi)\Gamma_0^3\left(\frac{2}{3} + \Gamma_0\sigma\right) - \Gamma^3\left(\frac{2}{3} + \Gamma\sigma\right)\right] \quad (16)$$

where σ is the ion diameter and the inverse screening length, Γ , is related to the inverse Debye screening length, κ , as

$$\Gamma(1 + \Gamma\sigma) = \frac{\kappa}{2} \quad (17)$$

where $\kappa^2 = \frac{4\pi l_b}{v}(1 - \phi)$ for monovalent ions. Here, $l_b = e^2/(4\pi\epsilon_0\epsilon_r k_B T)$, is the Bjerrum length which characterizes the distance where the electrostatic interaction between two elementary charges in a medium equals the thermal energy. The constant e is the elementary charge, ϵ_0 is the vacuum permittivity and ϵ_r is the relative static permittivity of the medium. Γ_0 is the screening length of a pure ionic liquid. There is some debate in the literature as to what the relative static permittivity should depend on. For most pure solvents, the relative static permittivity is expected to decrease with increasing temperature. This inverse relationship often results in an increase in the Bjerrum length, and therefore the strength of electrostatic interactions. However, it is also known that the introduction of charged species to the medium can also lead to a decrement in the relative static permittivity.^{58–60} A true first-principles theory would have to account for both these effects, which is clearly beyond the scope of this work. The value of the relative dielectric permittivity used in the model should best be interpreted as representing some average value across the range of compositions and temperatures considered.

We note that comparable behavior can be captured using the full Debye–Hückel free energy, where the two treatments

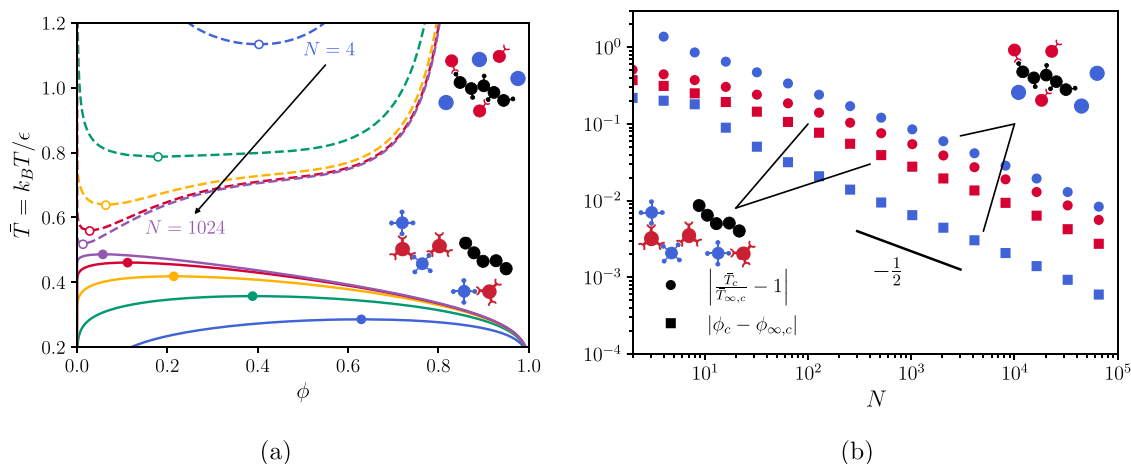


Figure 1. Phase behavior (a) and critical scaling (b) of a polymer + ionic liquid mixture only including the effect of clustering. The solid lines (a) and blue symbols (b) are the results obtained using clustering where $\epsilon_{cl} = 1.0$, $\nu_{cl} = 10^{-1}$ and $m_+ = 3$, $m_- = 4$. The dashed lines (a) and red symbols (b) are the results obtained using cation-polymer binding where $\epsilon_b = 1.0$, $\nu_b = 1.0$ and $m_+ = m_p = 1$, with an athermal $\chi = 1.95$. Square symbols in (b) correspond to the critical volume fraction of the polymer while the circular symbols correspond to the critical temperature.

converge in the dilute limit. Specifically, the Debye–Hückel limiting law is recovered by taking the point-charge limit ($\sigma \rightarrow 0$), which yields $f_{ele} \rightarrow -\kappa^3/(12\pi)$. This corresponds to a Voorn–Overbeek-like treatment of charged systems.⁶¹ However, in contrast to models incorporating finite-sized ions, the Debye–Hückel limiting law permits a divergence of the electrostatic free energy with increasing ion concentration. This behavior is unphysical at high packing fractions, where excluded volume effects limit ion overlap and lead to saturation of electrostatic contributions. Therefore, for dense systems such as polymer–ionic liquid mixtures, the use of MSA is essential for a realistic description.

3. PHASE EQUILIBRIUM AND CRITICAL POINT

In the previous sections, all equations were defined in terms of the polymer volume fraction, ϕ , where the composition of each ion can be obtained as $\phi_- = \phi_+ = \frac{1}{2}(1 - \phi)$, as dictated by electroneutrality.

At a fixed temperature, the condition for coexistence between phases α and β is given by

$$\left. \frac{\partial \Delta f_{\text{mix}}}{\partial \phi} \right|_{\alpha} = \left. \frac{\partial \Delta f_{\text{mix}}}{\partial \phi} \right|_{\beta} \quad (18)$$

$$\left(\Delta f_{\text{mix}} - \phi \frac{\partial \Delta f_{\text{mix}}}{\partial \phi} \right) \Big|_{\alpha} = \left(\Delta f_{\text{mix}} - \phi \frac{\partial \Delta f_{\text{mix}}}{\partial \phi} \right) \Big|_{\beta} \quad (19)$$

The first condition ensures equality of (exchange) chemical potential, while the second ensures equality of osmotic pressure. We note that, when the critical point is near $\phi \approx 1$, it is sometimes best to solve eqs 18 and 19 in terms of $1 - \phi$ rather than ϕ . Additional numerical stability can be achieved by solving for the logarithm of the compositions instead.

The critical point of the system is obtained from the condition

$$\frac{\partial^2 \Delta f_{\text{mix}}}{\partial \phi^2} = 0 \quad (20)$$

$$\frac{\partial^3 \Delta f_{\text{mix}}}{\partial \phi^3} = 0 \quad (21)$$

which is solved simultaneously to determine the critical composition ϕ_c and temperature T_c . All thermodynamic derivatives in this work are evaluated numerically using ForwardDiff.jl,⁶² a Julia package that implements forward-mode automatic differentiation.

4. KEY PHYSICAL EFFECTS

We shall now consider the effect of each interaction introduced in the previous section on the phase behavior of a polymer–ionic liquid mixture, before proposing a minimal model to reproduce the experimental results.

4.1. Clustering. We begin by examining the role of clustering, as shown by the solid curves in Figure 1, where temperature is normalized by the binding energy. In these calculations, we set $m_+ = 3$ and $m_- = 4$, corresponding to the number of putative binding sites on EMIM⁺ and BF₄[−], respectively.⁴⁹ By itself, clustering will favor demixing as it acts to reduce the translational entropy of the solvent. For sufficiently strong binding strength, clustering can induce phase separation if both m_+ and m_- exceed two. A more detailed discussion of this behavior is provided in the Supporting Information.

The resulting phase behavior exhibits upper critical solution temperature (UCST) behavior. As shown in Figure 1a, the critical point initially lies near polymer-rich compositions but shifts toward polymer-dilute regions with increasing chain length. White and Lipson³² previously argued that clustering increases the effective size of ionic species, thereby reducing the size asymmetry with the polymer and shifting the critical point toward polymer-rich compositions. This trend is partially reflected in our results. However, at sufficiently large polymer chain lengths, the critical point shifts back toward the polymer-dilute region, consistent with Flory–Huggins theory for mixtures with unfavorable polymer–solvent interactions. Indeed, this model recovers the classic Flory–Huggins scaling of the critical point with chain length, $N^{-1/2}$, as shown by the red markers in Figure 1b.

These trends can be rationalized by examining the average cluster size, shown in Figure 2. In pure ionic systems, cluster size is determined by the number of binding sites per ion and the binding strength (via eq 9). For example, at low temperatures, systems with $m_+ = m_- = 2$ form moderately large clusters, and systems with $m_+ = 3$ and $m_- = 4$ exhibit a divergence in cluster

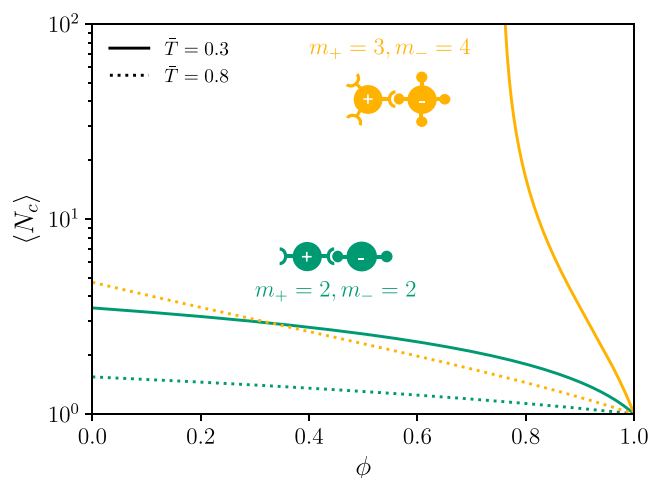


Figure 2. Average cluster size, obtained using eq 10, in a polymer–ionic liquid mixture varying number of sites on the ions (shown through the illustrations) at $\bar{T} = 0.3$ (solid line) and $\bar{T} = 0.8$ (dotted line). These results are independent from the chain length of the polymer. Parameters for clustering are identical to those used in Figure 1.

size with decreasing polymer volume fractions, indicating the onset of percolation. In both cases, increasing either the polymer volume fraction, disrupting cluster formation, or the temperature, weakening the binding interactions, leads to a reduction in average cluster size. This behavior is independent of polymer chain length.

At low polymer chain lengths, the mixing entropy is relatively large, resulting in a decreased critical temperature. Under these conditions, clustering plays a more significant role in the free energy of mixing, as seen from the solid red curve in Figure 3.

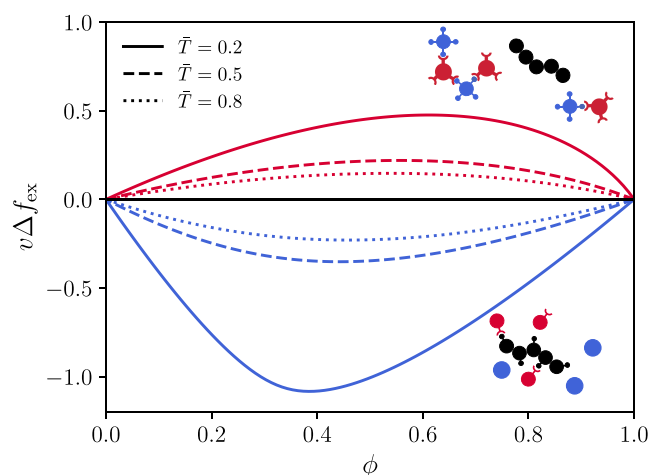


Figure 3. Contribution of clustering (red curves) and cation–polymer binding (blue curves) to the free energy of mixing for a polymer + ionic liquid system at various temperatures, with parameters corresponding to those used in Figure 1.

Here, clustering effectively acts as a correction to the entropy of mixing, accounting for the increased effective size of the ionic species. This correction is skewed toward polymer-rich compositions, leading to a shift in the critical point toward higher polymer concentrations.

As the polymer chain length increases, the mixing entropy decreases, leading to a higher critical temperature. At these higher temperatures, clusters are both smaller and more easily

disrupted, reducing their contribution to the free energy of mixing. This is reflected in the dashed and dotted red curves in Figure 3, which show a smaller and more symmetric excess contribution, resembling that of the standard Flory–Huggins interaction. As a result, the critical point shifts toward polymer-dilute compositions, recovering the characteristic $N^{-1/2}$ scaling behavior of Flory–Huggins theory, as shown in Figure 1b.

These results suggest that clustering among ionic liquids alone is insufficient to explain the experimentally observed phase behavior of polymer–ionic liquid systems.

4.2. Polymer–Cation Binding. An important interaction in IL–polymer mixtures is the reversible binding between cations and the polymer backbone. Lee et al.²² proposed, and Choi et al.³⁰ confirmed using molecular dynamics simulations, that cations such as EMIM⁺ can form hydrogen bonds with the ether oxygens in PEO via the electropositive hydrogens on the imidazolium ring. We model this interaction using eq 13, where, because $X^+, X^p \leq 1$, the binding contribution to the free energy of mixing is always negative. This indicates that polymer–cation binding enhances miscibility as illustrated by the blue curves in Figure 3. Note that, as temperature decreases, the minimum of the binding contribution shifts toward $\phi = \frac{1}{3}$, corresponding to the case where the number of cations equals the number of binding sites on the polymer.

To recover the experimentally observed LCST behavior, we can introduce an athermal, positive Flory–Huggins interaction parameter χ to account for the unfavorable mixing between hydrophilic polymers and hydrophobic ionic liquid moieties. At high temperatures, binding is weak and unfavorable χ interactions dominate, driving phase separation. Because this mechanism does not involve clustering or aggregation, the LCST is shifted toward the polymer-dilute compositions, and the critical scaling follows the Flory–Huggins prediction of $N^{-1/2}$, as shown by the red markers in Figure 1b. Moreover, the binding contribution, being skewed toward polymer-dilute compositions (Figure 3), may actually enhance this shift at lower chain lengths.

While the inclusion of explicit polymer–cation binding offers mechanistic insight and a more physically grounded treatment, the same qualitative features, particularly LCST-type demixing, can be captured using a temperature-dependent χ parameter. In this sense, the binding model serves to justify such a dependence. Indeed, the binding free energy can be algebraically rewritten in a form resembling eq 5, with effective parameters $A \sim -\log v_b$ and $B \sim -\epsilon_b$, highlighting the connection between reversible binding and the phenomenological χ parameter.

4.3. Electrostatic Correlations. In modeling ionic liquids, many prior studies have neglected electrostatic interactions,^{32,36,37} often arguing that ion pairing and clustering effectively render the system neutral, allowing the cation–anion pair to be treated as a single, uncharged species. However, as discussed in the previous section, clusters are dynamic entities. Their size and stability depend strongly on temperature and concentration, and under many conditions, ionic liquids predominantly consist of dissociated ions. Consequently, we argue that electrostatic interactions are a fundamental feature of ionic liquids and must be retained in any physically consistent model. In this work, we account for these interactions using the mean spherical approximation (MSA), as detailed in Section 2.3. Electrostatic correlations manifest as effective attractions between the ions that are inherently many-body in nature. Such effective attractions are disrupted when diluted by another

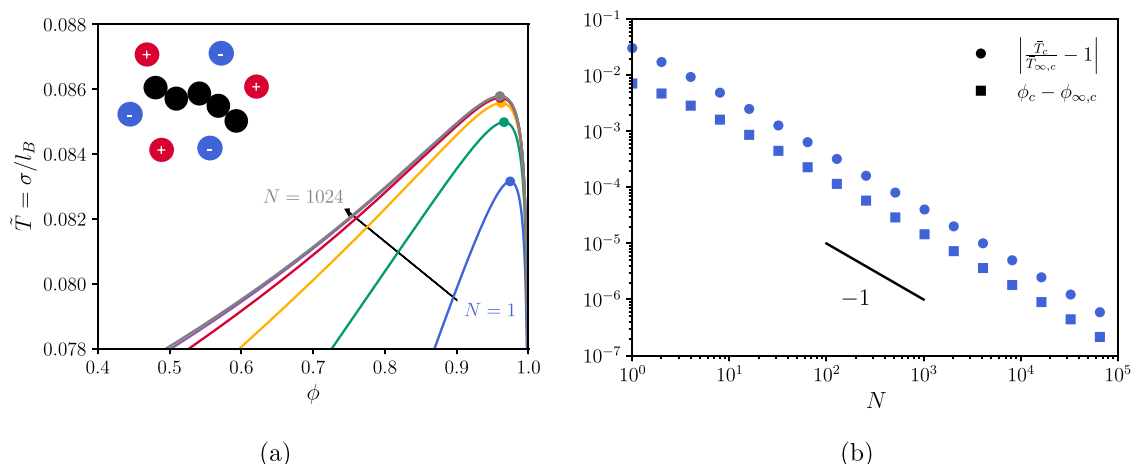


Figure 4. Phase behavior (a) and critical scaling (b) of a polymer–ionic liquid mixture only including electrostatic interactions. Square symbols in (b) correspond to the critical volume fraction of the polymer while the circular symbols correspond to the critical temperature.

neutral solvent, including polymeric solvent. This is evident by the positive contribution to the mixing free energy shown in Figure 5, providing the driving force for phase separation, even in the absence of other energetic interactions.^{63,64}

Figure 4 shows the phase diagrams for the polymer–IL mixture as several chain lengths with only electrostatic interactions, where the temperature axis is normalized using the Bjerrum length. For reference, assuming a dielectric constant $\epsilon_r \approx 6 - 15$ at $T = 298.15$ K for PEO, this corresponds to a reduced temperature range of $\tilde{T} \approx 0.04 - 0.10$, with $\sigma = 0.4$ nm.

Due to the effective repulsion between the polymer and the ionic species, as the polymer disrupts favorable electrostatic interactions, a polymer–IL mixture with only electrostatic interactions exhibits UCST behavior. However, unlike classical Flory–Huggins systems, the critical concentration remains consistently skewed toward polymer-rich compositions, even at large chain lengths, and approaches a finite value as $N \rightarrow \infty$. These features contrast sharply with the behavior observed in Figure 1, and is consistent with known trends in charged systems,⁶⁴ including polyelectrolytes.^{46,65–67} To understand this shift, we consider the scaling of the inverse screening length, which follows

$$\Gamma \sim (1 - \phi)^{1/2} \quad (22)$$

at low ionic strengths. As such, Γ decreases sharply as $\phi \rightarrow 1$, resulting in weaker electrostatic interactions at high polymer concentrations. This asymmetry is reflected in the electrostatic contribution to the free energy of mixing (Figure 5), which is positive and skewed toward polymer-rich compositions. Importantly, this skewness persists even at elevated temperatures, with the maximum in the excess free energy occurring around $\phi \approx 0.65$. Note that when the mixing entropy is included, the combined free energy produces a composition of the polymer-rich phase at coexistence close to $\phi \approx 1$; see Figure S1.

Although increasing chain length modifies the entropy of mixing and introduces a competing asymmetry, this contribution diminishes with increasing N . As a result, the electrostatic contribution dominates, and the critical concentration remains near the polymer-rich region, as shown in Figure 4a.

An additional noteworthy result is the scaling of the critical parameters, shown in Figure 4b. We find that both the critical temperature and concentration scale as N^{-1} , in contrast to the $N^{-1/2}$ scaling predicted by Flory–Huggins theory. To our

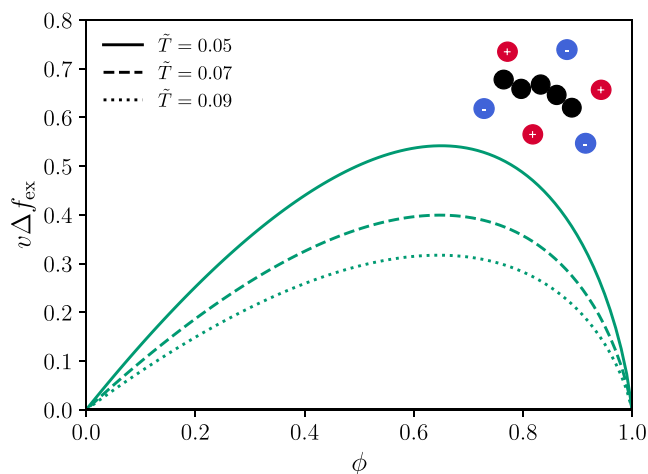


Figure 5. Electrostatic contribution to the free energy of mixing for a polymer–ionic liquid system at various temperatures.

knowledge, this scaling has not been reported in experiments or simulations. However, this is not entirely surprising. The experiments of Lee et al.,²² for example, involve multiple competing interactions, making it impossible to isolate the effects of electrostatics. Furthermore, our result is derived using a reduced temperature based on the Bjerrum length, which incorporates the dielectric constant. As ϵ_r typically decreases with increasing temperature, direct experimental validation of this scaling will be challenging. However, simulations could be designed to isolate electrostatic effects and verify this scaling. We note that a version of the theory presented in section 2.3 which incorporate a temperature-dependent dielectric constant has shown that this decrease in dielectric constant can induce LCST behavior by itself.⁴⁸

Another striking finding from our analysis is the behavior of the inverse screening length at the critical point, which, when extrapolated numerically to infinite chain length, converges to a remarkably simple value

$$\lim_{N \rightarrow \infty} (\Gamma_c \sigma)^2 \approx \frac{1}{2} \quad (23)$$

This result is of similar magnitude to the limit predicted by the restricted primitive model studied by Luijten et al.,⁶⁴ though our value is slightly smaller. We hypothesize that this limiting

behavior may reflect optimal ion packing in the condensed phase under strong electrostatic correlations. A detailed exploration of this hypothesis is beyond the scope of this work and is left for future investigation.

More broadly, while the mean spherical approximation (MSA) captures long-range electrostatic interactions, it has also been argued that it implicitly accounts for short-range clustering.^{68,69} This interpretation has important modeling implications: combining an explicit clustering free energy with an MSA-based electrostatic term risks double counting the same physical effect. Since the MSA approach alone reproduces the experimentally observed shift in critical concentration (Figure 4), we contend that it provides a more parsimonious and physically grounded basis for modeling ionic liquid-polymer mixtures. For cases where a more explicit treatment of ion pairing is desired, extensions such as binding-MSA^{70,71} may offer a systematic and rigorous framework, though less interpretable.

5. MINIMAL MODEL

Building on the insights developed in the preceding sections, we now introduce a minimal model for polymer-ionic liquid mixtures.

If one were to attempt rigorously accounting for all interactions present in a polymer-IL system, it would require combining clustering, polymer-cation binding, electrostatic and χ interactions. Aside from introducing significant model complexity, these contributions are all temperature-dependent and can become intertwined, leading to parameter degeneracy and ambiguity in interpretation.⁷² An example of this has already been hinted at in section 4.2 where the effects of polymer-cation binding could be convoluted with a temperature-dependent χ parameter. For these reasons, a more pragmatic and robust modeling strategy may be to omit polymer-cation binding altogether and absorb their effects into an temperature-dependent χ with both enthalpic and entropic contributions (leading to an LCST). As electrostatic correlations cannot be captured by an effective χ parameter, and implicitly account for clustering, the MSA contribution will be retained where we set $\sigma = 0.4$ nm, as we have done in our prior work.⁴⁸

The resulting expression is given by

$$\Delta f_{\text{mix}} = \frac{1}{v} \left[\frac{\phi}{N} \log \phi + (1 - \phi) \log(1 - \phi) + \chi \phi(1 - \phi) \right] + \frac{1}{\pi} \left[(1 - \phi) \Gamma_0^3 \left(\frac{2}{3} + \Gamma_0 \sigma \right) - \Gamma^3 \left(\frac{2}{3} + \Gamma \sigma \right) \right] \quad (24)$$

where each term, its parameters and influence on the free energy of mixing has been discussed in the previous section. This expression only departs slightly from the classical Flory-Huggins theory, but retains the key physical ingredients necessary to reproduce the thermodynamic behavior of these systems. Specifically, we propose that the mixing free energy requires only three contributions: entropy of mixing, polymer-ion interactions (captured via a temperature-dependent Flory-Huggins parameter given by eq 5), and electrostatics treated using the mean spherical approximation (MSA). To avoid overfitting, an all-too-common pitfall,^{72,73} we deliberately restrict ourselves to just four adjustable parameters: the molecular weight per effective monomer, the coefficients A and B in χ between the polymer and cation, and a single dielectric constant.

For the purposes of this section, we will focus solely on the PEO-[EMIM][BF₄] system studied by Lee et al.²² We have applied our model to two other polymer-ionic liquid mixtures,²⁶ with the results shown in the Supporting Information. The resulting phase diagram is shown in Figure 6, with the fitted values of the parameters given in Table 1. It is

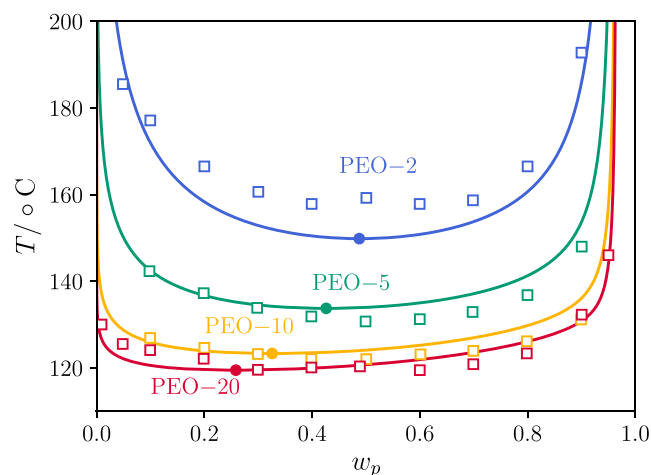


Figure 6. Phase diagram for PEO+[EMIM][BF₄] at various PEO chain lengths (color) where solid curves represent fittings obtained using a combination of MSA + χ contributions. Model parameters are given in Table 1. w_p is the weight fraction of the polymer. Data are taken from ref 22, where i in the notation PEO- i denotes the approximate kg/mol of the polymer chain.

Table 1. Parameters Corresponding to the PEO + [EMIM][BF₄] System for the Approach Using a Combination of the MSA + χ Contribution

parameter	values	units
M_w/N	237.2	g/mol
A	2.509	-
B	-923.3	K
ϵ_r	9.228	-

important to note that fitting is only performed using data from the PEO-20 data set. The results given in Figure 6 for other chain lengths are completely predictive. The close agreement between the theoretical predictions and the experimental data highlights the model's extrapolative power and physical grounding, achieved using a minimal set of well-motivated parameters. We considered fitting on the PEO-2 data set but found that the parameter extrapolated poorly, likely due to the fact that shorter PEO chains are expected to have different dielectric constants⁷⁴ (these results are shown in the SI). We also note that, as discussed by Lee et al.,²² large uncertainties are to be expected in the measurements of cloud points. As such, small disagreements between predictions and experimental data, such as that for PEO-2 and PEO-5 mixtures, can be partially attributed to this.

We now briefly discuss the parameter values obtained from our model. If we consider the polymer chain length predicted in Table 1 and our prescribed value for σ , it suggests that each coarse-grained segment corresponds to approximately five PEO monomers, which is consistent with the large Kuhn length of PEO.⁷⁵ Note that while M_w/N is expected to be degenerate with σ , as both quantify the size of the chain, it is expected that this correspondence should remain the same.

The fitted interaction parameters between the ions and the polymer are similarly reasonable. They are intended to reflect the temperature-dependent balance between favorable ion–polymer binding at low temperatures and unfavorable hydrophilic–hydrophobic interactions at higher temperatures. The magnitudes are comparable to those used for systems with similar lower critical solution temperature (LCST) behavior, such as water + PEO mixtures.^{54,55} The condition $A > 0$ and $B < 0$ induces the LCST, as observed in Figure 6.

Crucially, the inclusion of electrostatic interactions is necessary to capture the observed shift in critical concentration. Without them, the coexistence curve would be skewed toward polymer-dilute compositions. This arises from the inherent skew in the electrostatic contribution compensating for the skew present in the entropy of mixing (see SI for example figures), as discussed earlier. The fitted dielectric constant falls between the known values for PEO (5–10)⁷⁴ and for common ionic liquids (5–20),⁷⁶ lending further physical credibility to the model. Note that this value is expected to depend on the specific polymer–ionic liquid mixture. We also assume the dielectric constant is independent of temperature, although it is known to generally decrease with increasing temperature. Therefore, we interpret this value as an effective average over the temperature range considered. We note that, in our previous work,⁴⁸ the inclusion of this dielectric decrement by itself can induce LCST behavior in aqueous polyelectrolytes. In polymer + IL systems, even if the decrement in relative static permittivity with increasing temperature had been captured explicitly, due to the already low value in the dielectric constant, this would be insufficient to drive such behavior.

An interesting feature in Figure 6 is that the critical composition shifts toward polymer-dilute regions with increasing chain length. Experimentally, while Lee et al.²² did not explicitly report the critical point, we note that the experimental coexistence curves near criticality are relatively flat, making the precise location of the critical point difficult to extract. Within this model, the trend arises because, for the two-phase region to close at lower temperatures, the favorable χ interactions must eventually outweigh the electrostatic contributions. Consequently, the critical behavior becomes dominated by the entropy of mixing, which naturally drives the critical point toward the polymer-dilute limit as chain length increases. However, if the dielectric decrement with temperature is sufficiently strong so as to lead to an LCST driven by electrostatic correlations alone, then based on the limited shift in critical composition in Figure 4a, we would observe minimal shifts in critical composition as the chain length increases.

Overall, the results presented in Figure 6 demonstrate that even a system as complex as PEO + [EMIM][BF₄], involving a convolution of electrostatic and clustering interactions, can be accurately and transparently modeled using a minimal set of physically motivated parameters.

6. CONCLUSION

In this work, we have developed a physically motivated framework for modeling the phase behavior of polymer–ionic liquid mixtures, with particular emphasis on understanding the role of clustering, polymer–cation binding and, electrostatics correlations. Starting from Flory–Huggins theory, we incorporated key nonideal effects through excess contribution to the mixing free energy expression. Through systematic analysis of each contribution, we clarified the distinct and competing roles they play in determining the phase behavior.

Our results demonstrate that although clustering leads to phase separation which skews slightly toward the polymer-rich compositions, it cannot by itself account for the experimentally observed lower critical solution temperature (LCST) behavior in systems. Conversely, explicit modeling of cation–polymer binding, combined with an unfavorable entropic Flory–Huggins interaction, can capture the LCST behavior and yields critical scaling consistent with Flory–Huggins predictions. However, this effect alone does not reproduce the shape of the coexistence curve with increasing molecular weight observed experimentally in polymer–ionic liquid systems.

Importantly, we find that electrostatic interactions, often neglected in prior treatments, have a profound and persistent influence on the phase behavior. The MSA treatment captures not only the many-body, long-range nature of electrostatic interactions but also implicitly reflects the tendency of ions to form clusters, eliminating the need for a separate clustering term. The inclusion of this contribution leads to a complete transposition of the critical concentration toward the polymer-rich compositions, something that cannot be explained using a simple Flory–Huggins-type theory. Interestingly, the critical concentration now approaches a finite value close to $\phi = 1$, in contrast to the typical result from Flory–Huggins theory. The change in the critical temperature and concentration (from the infinite chain-length limit) both scale as N^{-1} , as opposed to the $N^{-1/2}$ scaling from the Flory–Huggins theory.

Finally, we propose a minimal model that combines only three ingredients (mixing entropy, temperature-dependent χ interactions, and electrostatics) yet reproduces key features of experimental phase behavior using a minimal set of physically interpretable parameters. This model highlights how a carefully constrained theoretical framework can yield quantitative predictions while avoiding the pitfalls of overfitting and parameter degeneracy. Good agreement with experimental data on several polymer–ionic liquid mixtures demonstrates the broad applicability of this model.

Together, these findings offer a unified and extensible framework for the thermodynamic modeling of polymer–ionic liquid systems, with implications for both fundamental understanding and materials design.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.5c04858>.

Derivation of the clustering and binding terms, and supplementary figures showing the total mixing free energy for electrostatic and polymer–cation binding systems (PDF)

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Notes

The code used to generate the results has been made available open-source using Clapeyron.jl⁷⁷ in the following GitHub repository: <https://github.com/zgw-group/PEO-IL/tree/main>.

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research is supported by the Hong Kong Quantum AI Lab, AIR@InnoHK of the Hong Kong Government. Special thanks to Jonathan Aalto for initiating this work as part of his rotation. P.J.W. would like to thank Dr. Samuel Varner and Prof. Alejandro Gallegos for insightful discussions.

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